



Roll No:

BPHARM
(SEM III) THEORY EXAMINATION 2025-26
PHYSICAL PHARMACEUTICS I

TIME: 3 HRS

M.MARKS: 75

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

10 x 2 = 20

1. Attempt all questions in brief.

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| a. | Define ideal solubility parameter and give its significance. |
| b. | Define Solvation, Desolvation, and Association with one example each. |
| c. | Explain critical solution temperature and mention its types. |
| d. | Define eutectic mixture. Give two pharmaceutical examples. |
| e. | Define Sorensen's pH scale and write its mathematical expression. |
| f. | What is Critical Micellar Concentration (CMC)? Mention any two factors affecting it. |
| g. | Define Detergency. How does it differ from cleaning? |
| h. | Differentiate between Nernst potential and Zeta potential. |
| i. | What are chelating agents? Write any two pharmaceutical applications. |
| j. | Define viscosity and mention one factor affecting the viscosity of liquid. |

SECTION B

2 x 10 = 20

2. Attempt any two parts of the following:

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| a. | Describe in detail the solubility of liquids in liquids. Discuss the types, factors affecting mutual solubility, CST, and pharmaceutical applications. |
| b. | Derive Raoult's Law for ideal solutions. Discuss positive and negative deviations from Raoult's law with suitable examples and diagrams. |
| c. | Explain various methods used for the determination of surface and interfacial tension: |

SECTION C

7 x 5 = 35

3. Attempt any five parts of the following:

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| a. | Define isotonicity. Differentiate between isosmotic and isotonic solutions. Explain methods used to adjust tonicity |
| b. | Describe classification, properties, and pharmaceutical applications of complexation. Add examples for each class. |
| c. | Classify surface-active pharmaceutical systems. Explain HLB system and its role in emulsification and solubilization. |
| d. | Explain buffer action. Describe Henderson-Hasselbalch equation and applications of buffers in biological and pharmaceutical systems. |
| e. | Differentiate between crystalline and amorphous states based on structure, melting point, solubility, and stability. Give examples. |
| f. | Discuss Nernst Distribution Law. Explain its applications and limitations in pharmacy. |
| g. | Explain protein binding, types of protein-drug interactions, factors affecting protein binding, and its significance in pharmacokinetics. |